



Temitope Oriowo

Bioinformatics Analyst

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SUMMARY

Bioinformatician with 5+ years of experience in high-throughput sequencing data analysis and computational genomics, with a strong focus on variant analysis and statistical genetics. I develop scalable, reproducible pipelines for bulk and genome-scale sequencing data, including variant calling, and downstream SNP-based analyses. My work includes GWAS implementation and the analysis of population-scale genomic variation, providing a solid foundation for polygenic score (PGS) modelling and genetic risk prediction. I am highly proficient in Python, R, and Unix/Linux, with advanced workflow development using Snakemake and Nextflow in high-performance computing environments.

WORK EXPERIENCE

01/05/2022 — current

Doctoral Researcher (Bioinformatics),

Leibniz Institute for the Analysis of Biodiversity Change (Museum Koenig), Bonn, Germany

- Developed and optimized scalable, reproducible bioinformatics pipelines for whole-genome sequencing data using Snakemake, Python, R, and Bash, including variant calling, filtering, and annotation.
- Performed SNP-based analyses on population-scale genomic datasets, including variant extraction, filtering, and downstream analyses using tools such as bcftools, PLINK, and custom scripts.
- Implemented GWAS workflows (e.g., GEMMA and k-mer-based GWAS approaches such as kmerGWAS) to analyse genotype–phenotype associations, providing a strong foundation in statistical genetics and variant-based modelling
- Deployed and scaled workflows in HPC environments (SLURM, SGE) using Docker and Singularity, ensuring reproducibility and computational efficiency

EDUCATION

01/05/2022 — 30/06/2026

PhD in Biology (Evolutionary Biology),

University of Bonn, Evolutionary Biology, Bonn, Germany

Thesis: Genomic differentiation, hybridization, and species delimitation in *Phoxinus*

Focus: population genomics, variant analysis, statistical modelling, and bioinformatics pipeline development

12/08/2019 — 20/09/2021

Master of Science (Animal Breeding and Genetics)

Wageningen University & Research / University of Natural Resources and Life Sciences, Vienna (BOKU)

04/08/2012 — 18/07/2017

Bachelor of Agriculture (Animal Sciences),

Bowen University, Iwo, Nigeria

CORE SKILLS

- **Variant and integrative genomic analysis**
SNP discovery, filtering, and interpretation; GWAS (GEMMA, k-mer-based approaches), population genomics, and variant-based modelling
- **Next-generation sequencing (NGS) data analysis**
Whole-genome sequencing (WGS), variant calling workflows, VCF processing, genome assembly, and annotation
- **Programming for genomic data analysis**
Python, R, SQL, and Bash for scalable data processing, statistical analysis, and workflow automation
- **High-performance computing and reproducible workflows**
Linux environments, SLURM/SGE, Docker/Singularity, and modular pipeline development (Snakemake, Nextflow)
- **Scientific reporting and interdisciplinary collaboration**
Integration of genomic and phenotypic data, reproducible reporting, and publication-quality visualization

LANGUAGE

- English (C2 level)
- German (A2 level)

PUBLICATIONS

Oriowo, T.O., Smith, S., Thormann, J., Sternberg, N., & Boehne, A. (2026). Different sex determination systems in two closely related Eurasian minnow (Phoxinus) species. *Heredity*. <https://doi.org/10.1038/s41437-026-00827-8>.

Oriowo, T. O., Chrysostomakis, I., Martin, S., Kukowka, S., Brown, T., Winkler, S., & Stange, M. (2025). A chromosome-level, haplotype-resolved genome assembly and annotation for the Eurasian minnow (Phoxinus phoxinus). *GigaScience*, 14, giae116. <https://doi.org/10.1093/gigascience/giae116>

Macphee, A., **Oriowo, T. O.**, Sternberg, N., & Stange, M. (2025). Comparison of whole-genome sequencing performance from fish swabs and fin clips. *BMC Research Notes*, 18(1), 15. <https://doi.org/10.1186/s13104-024-06536-3>

Sternberg, N., Bodenheimer, A., **Oriowo, T. O.**, Podsiadlowski, L., & Stange, M. (2025). Human impacts on the distribution and genetic diversity of Eurasian minnows in the Rhenish Massif. *Knowledge & Management of Aquatic Ecosystems*, 426, 20. <https://doi.org/10.1051/kmae/2025015>

LINKS

Linkedin: [linkedin.com/in/temitope-oriowo](https://www.linkedin.com/in/temitope-oriowo)

Github: <https://github.com/Temmy-bioinformatics>